

PFC–Limbic Flexibility: A Conceptual Framework and Hybrid EEG–fMRI Research Agenda

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Abstract

Contemporary affective and cognitive neuroscience increasingly frames cognition and emotion as emergent properties of dynamic, context-dependent interactions among distributed brain networks. In this narrative, concept-driven review, we synthesize representative evidence on prefrontal–limbic coordination supporting emotion regulation, decision-making, and social behavior, and we introduce PFC–limbic flexibility as a bridging construct. We define PFC–limbic flexibility as the capacity of prefrontal and cingulate systems to adjust the strength and timing of interactions with amygdala–hippocampal and valuation circuits in a context-sensitive manner; claims about causal direction are treated as theoretical unless supported by effective-connectivity models. We integrate network-neuroscience perspectives, meta-analytic findings on task-modulated amygdala–prefrontal coupling during regulation, and allostasis-based accounts of stress-related plasticity in PFC and hippocampus. We also discuss interpersonal synchrony and digital emotion contagion as extended contexts of influence, distinguishing robust behavioral evidence from limited neural evidence and noting an editorial expression of concern associated with a prominent platform-scale experiment. In applied domains, we examine consumer neuroscience and EEG-based neuromarketing, emphasizing interpretability, statistical power, analytic flexibility, reproducibility, and dual-use risk. To operationalize PFC–limbic flexibility, we propose a multidimensional framework spanning (i) context sensitivity of connectivity, (ii)

configurational diversity of within-subnetwork states, and (iii) neural-behavioral coupling to regulation and resistance outcomes. Finally, we outline a staged EEG-only → fMRI-only → simultaneous EEG-fMRI agenda for testing persuasion-related hypotheses under explicit transparency and governance constraints.

Keywords: prefrontal cortex; amygdala; limbic system; emotion regulation; functional connectivity; allostasis; EEG-fMRI; consumer neuroscience; reproducibility; dual-use

1. Introduction

Classical cognitive theories often treated cognition and emotion as separable faculties with distinct neural substrates. Across lesion work, systems neuroscience, and computational network modeling, this dichotomy has become increasingly difficult to maintain. A growing consensus holds that cognitive-emotional functions arise from interacting, overlapping neural systems whose contributions depend on task demands, internal states, and broader network context.

Pessoa's "cognitive-emotional brain" framework argues that prefrontal cortex (PFC), anterior cingulate cortex (ACC), insula, amygdala, hippocampus, and related valuation circuits participate flexibly in diverse functions rather than belonging to strictly 'cognitive' or 'emotional' modules (Pessoa, 2013). Under this view, cognitive-emotional integration refers to the ongoing coordination between processes often labeled as cognitive—appraisal, working memory, cognitive control, and decision-making—and those labeled as emotional—salience tagging, threat detection, arousal, and value representation.

1.1. Gaps motivating the present framework

Despite extensive literatures on emotion regulation, stress, digital emotion, interpersonal synchrony, and consumer neuroscience, three gaps motivate the present review and agenda.

1. **Integration gap.** Findings on task-dependent amygdala-PFC coupling, stress-related plasticity in PFC/hippocampus, interpersonal synchrony, and applied persuasion are often studied in isolation. A unifying, subsystem-specific lens centered on PFC-limbic circuitry remains rare.
2. **Construct gap.** Adjacent constructs (cognitive flexibility, regulatory flexibility, dynamic functional connectivity) do not explicitly target a PFC-limbic subsystem nor articulate how stress and context jointly shape susceptibility or resistance to emotionally framed influence.
3. **Methods-and-governance gap.** Although hybrid EEG-fMRI techniques are technically mature, theory-driven paradigms for persuasion and resistance remain limited, and ethical governance has not been integrated as a first-class design constraint.

1.2. Aims and contributions

This paper has three aims. First, we synthesize representative evidence on neural mechanisms of cognitive-emotional integration, emphasizing PFC-limbic interactions. Second, we introduce PFC-limbic flexibility as a multidimensional construct intended to bridge basic affective neuroscience, social interaction, and applied persuasion while remaining falsifiable. Third, we outline a pragmatic hybrid

EEG–fMRI research agenda, explicitly staged to match laboratory feasibility and to reduce analytic flexibility and dual-use risk.

2. Scope and Approach

2.1. Nature of the review

This article is a narrative, integrative review rather than a PRISMA-style systematic review. The goal is conceptual integration and hypothesis generation, not exhaustive coverage. We curate findings that are particularly informative about (a) PFC–limbic circuits, (b) dynamic coordination and plasticity, and (c) the intersection of persuasion-related contexts with reproducibility and ethics.

2.2. Search strategy and selection criteria

Targeted searches were conducted for literature published primarily between 2010 and mid-2025 (with selective inclusion of foundational earlier work). Searches used combinations of terms including: cognitive–emotional integration; amygdala–prefrontal connectivity; emotion regulation PPI; allostasis; stress-induced plasticity; digital emotion contagion; consumer neuroscience; neuromarketing; simultaneous EEG–fMRI; artifact reduction; dynamic functional connectivity; and network flexibility.

We prioritized peer-reviewed journal articles, meta-analyses, and methodological guidance on power, transparency, and EEG–fMRI. We excluded non–peer-reviewed commentary and empirical reports without clear relevance to connectivity mechanisms or decision-related outcomes. Because search engines and indexing evolve over time, the exact set of retrieved records is not fully reproducible; the cited literature should be regarded as illustrative and theory-guiding rather than exhaustive.

2.3. Levels of evidence and interpretive stance

Because the review integrates domains with uneven evidentiary maturity, we adopt an explicit ‘levels of evidence’ stance. We treat meta-analytic findings and convergent multimethod results as higher weight; single-study findings, especially those from small samples or highly flexible analytic pipelines, are treated as hypothesis-generating. We explicitly label claims about digital exposure shaping PFC–limbic circuits as plausible hypotheses unless supported by longitudinal neuroimaging evidence.

3. Neural Mechanisms of Cognitive–Emotional Integration

3.1. Network perspectives on cognition–emotion interactions

Network-based accounts propose that cognitive–emotional functions emerge from interactions within large-scale systems rather than isolated ‘emotion’ or ‘cognition’ modules (Pessoa, 2013; Bassett & Sporns, 2017; Shine et al., 2016). Within this view, amygdala contributions are often framed as relevance or salience detection across domains, while PFC/ACC support valuation, conflict monitoring, and goal maintenance. Critically, these roles are context-dependent and can shift with internal state (e.g., stress) and task demands.

3.2. Amygdala–prefrontal dynamics in emotion regulation

Connectivity analyses have been especially informative about how PFC and amygdala coordinate during emotion regulation. Meta-analytic evidence from psychophysiological interaction (PPI) studies indicates reliable task-modulated coupling between amygdala and prefrontal regions during regulation tasks (Berboth & Morawetz, 2021). Importantly, these results do not support a simple one-direction ‘PFC inhibits amygdala’ narrative; the magnitude and sign of coupling vary with strategy (e.g., reinterpretation vs. distancing), task instructions (explicit vs. implicit), and individual differences in regulation success.

A methodological clarification is warranted: gPPI and related functional connectivity approaches quantify condition-dependent changes in correlation structure, but they do not, by themselves, establish causal directionality. Claims about top-down control should therefore be framed as theoretical interpretations unless supported by models aimed at effective connectivity (e.g., DCM) or convergent causal designs.

At a broader level, canonical models of cognitive emotion regulation highlight the role of cognitive control and reappraisal systems engaging lateral and medial PFC alongside modulation of affect-generative systems (Ochsner & Gross, 2005). Large meta-analyses of reappraisal neuroimaging studies support robust activation of control-related regions and modulation of bilateral amygdala during down-regulation, while also emphasizing heterogeneity across tasks and study designs (Buhle et al., 2014).

3.3. PFC–hippocampal circuits, stress, and allostasis

Allostasis describes achieving stability through change by adjusting internal parameters in response to environmental demands; allostatic load indexes the cumulative costs of chronic or dysregulated stress (McEwen & Gianaros, 2011). Stress-related neuroplasticity work indicates that prefrontal and hippocampal circuits are stress sensitive across the life course, with implications for working memory, prospective planning, and regulation (McEwen, 2013).

A complementary line of work links stress to a shift in behavioral control, where stress can reduce goal-directed action control and favor habitual responding. A review by Schwabe and Wolf (2011) synthesizes evidence that stress-related neuroendocrine processes may bias corticostriatal circuitry toward habits. This literature strengthens the plausibility that chronic stress could reduce the adaptability of PFC–limbic coordination relevant for resisting emotionally charged, short-term persuasive appeals.

3.4. Interpersonal synchrony and extended social contexts

Cognitive–emotional integration can also be examined at the interpersonal level. Hyperscanning studies (simultaneous recording from multiple individuals) suggest that coordinated neural dynamics may support empathy, cooperation, and shared attention. For example, EEG hyperscanning work reports increases in brain-to-brain synchronization during shared painful experiences and links such synchrony to empathy and prosocial outcomes (Peng et al., 2021).

The hyperscanning literature remains methodologically heterogeneous (tasks, metrics, and statistical models), so claims about general mechanisms should be made cautiously. Nevertheless, these studies

motivate a broader hypothesis: social interaction may involve coupling not only within brains but also across brains, potentially altering the context in which PFC–limbic regulation occurs.

3.5. Digital emotion contagion and platform-scale affective contexts

Digital platforms provide large-scale environments for affective influence. A widely cited experimental manipulation of Facebook News Feed content reported changes in the emotional tone of users' own posts consistent with emotional contagion (Kramer et al., 2014). However, the ethical handling of this study has been disputed, and PNAS published an editorial expression of concern noting concerns about informed consent and participant awareness (Verma, 2014).

Beyond single studies, reviews argue that platform architecture—algorithmic curation, engagement incentives, and virality—can produce self-reinforcing cycles that amplify particular affective states (Goldenberg & Gross, 2020). At present, direct neuroimaging evidence linking chronic digital exposure to specific changes in PFC–limbic circuit flexibility remains limited. Accordingly, we treat claims about neural remodeling from digital emotional environments as testable hypotheses rather than established effects.

4. PFC–Limbic Flexibility: A Bridging and Falsifiable Construct

4.1. Definition

We define PFC–limbic flexibility as the capacity of prefrontal and cingulate systems to adjust the strength, timing, and (when supported by appropriate models) directionality of functional interactions with limbic and hippocampal regions in response to changing internal states, external demands, and goals. This construct is intended to bridge (a) task-modulated amygdala–PFC coupling in emotion regulation, (b) stress-related plasticity affecting regulatory capacity, and (c) contextual susceptibility or resistance to emotionally framed persuasion.

4.2. Relation to adjacent constructs

PFC–limbic flexibility overlaps with, but is not reducible to, several existing constructs:

- Cognitive flexibility: typically operationalized as set shifting or task switching, often emphasizing dorsolateral PFC and behavior rather than limbic coupling.
- Regulatory flexibility: typically defined at the psychological level as choosing and switching emotion regulation strategies based on context.
- Dynamic functional connectivity (dFC): temporal variability of functional coupling across time or states, often whole-brain rather than subsystem-specific.

The proposed added value is (i) subsystem specificity (a defined PFC/ACC–amygdala/hippocampus–valuation core) and (ii) functional linkage to stress/allostasis and persuasion outcomes, with explicit operational definitions and falsifiability criteria.

4.3. Multidimensional operationalization

To make the construct empirically tractable, we propose a multidimensional structure spanning: (1) context sensitivity, (2) configurational diversity, and (3) neural-behavioral coupling. Table 1 provides a pragmatic operationalization matrix intended to standardize measurement and reduce analytic degrees of freedom (Hutchison et al., 2013; Preti et al., 2017; Shine et al., 2016).

Table 1. Operationalization matrix for PFC–limbic flexibility (pragmatic proposal).

Dimension	Operationalization candidates and guardrails
Context sensitivity Connectivity changes across conditions (e.g., passive vs regulation vs resistance)	fMRI: gPPI contrasts; condition-wise coupling changes within predefined PFC/ACC ↔ amygdala/hippocampus ROIs. EEG: Event-related LPP amplitude; frontal midline theta during regulation/resistance; attention indices (posterior alpha). Behavior: Downregulation success; persuasion ratings; incentive-compatible choice (short vs long-term). Reliability/guardrails: Pre-register ROIs and contrasts; report effect-size uncertainty; split-half/stability checks.
Configurational diversity Breadth of within-subnetwork states explored (avoid “locked-in” patterns)	fMRI: Subnetwork dFC state counts; entropy of state occupancy; within-subnetwork modularity/participation coefficients. EEG: EEG microstates (exploratory); time–frequency state clustering aligned to message epochs. Behavior: Behavioral flexibility across trials; strategy switching; reduced perseveration. Reliability/guardrails: Avoid post-hoc window fishing; sensitivity analyses for window length; cross-validated state models.
Neural-behavioral coupling Predictive linkage of neural indices to behavior and regulation outcomes	fMRI: Trial-wise mixed models linking connectivity indices to ratings/choices; mediation models with stress as moderator. EEG: EEG-behavior coupling (theta/LPP) and EEG-informed fMRI regressors. Behavior: Resistance to persuasion; delayed-choice consistency; memory for arguments/counterarguments. Reliability/guardrails: Out-of-sample prediction (nested CV); correction for multiple tests; transparent analytic pipeline.

4.4. Falsifiability and boundary conditions

The construct is intended to be falsifiable. If proposed indices (context sensitivity, configurational diversity, and neural-behavioral coupling) do not show convergent validity across tasks/modality combinations, do not demonstrate acceptable reliability, or fail to predict regulation and resistance outcomes above baseline behavioral models, then PFC–limbic flexibility should be revised or

abandoned. Boundary conditions likely include task ecological validity, stress intensity/timing, and individual differences (e.g., trait anxiety) that may alter circuit engagement.

5. Applied Context: Persuasion, Consumer Neuroscience, and Neuromarketing

5.1. Valuation circuits and consumer choice

Consumer neuroscience studies how the brain encodes value, preference, and choice. Critical reviews highlight a valuation network involving vmPFC/OFC, ventral striatum, and hippocampal contributions to remembered value and contextual learning (Plassmann et al., 2012; Hsu & Yoon, 2015). These value signals integrate affective responses, social information, and cognitive goals rather than reflecting purely 'economic' computations.

5.2. EEG-based neuromarketing: promise and limitations

EEG has become popular in applied neuromarketing due to cost and temporal resolution. A recent systematic review summarizes common features used to index attention, engagement, and memory (e.g., frontal alpha asymmetry, theta power, ERPs) but also highlights methodological heterogeneity and interpretability risks (Khondakar et al., 2024).

Persistent limitations include small samples and unstable effect estimates (Button et al., 2013), reverse inference from coarse EEG signatures to complex constructs (e.g., 'true preference'), and proprietary pipelines that impede reproducibility. These limitations are particularly salient when neuromarketing claims are used to justify targeted influence in commercial or political contexts (Ariely & Berns, 2010).

5.3. PFC–limbic flexibility hypotheses in consumer and persuasion contexts

Viewing persuasion through the lens of PFC–limbic flexibility motivates testable hypotheses rather than immediate applications. For example:

- Individuals with higher context-sensitive PFC–amygdala and PFC–hippocampal modulation may show greater capacity to resist high-arousal appeals in favor of long-term goals.
- Chronic stress/allostatic load may reduce flexibility, biasing decisions toward reactive, short-term valuation.
- Multimodal indices combining EEG markers of cognitive control (e.g., frontal midline theta) with fMRI connectivity may outperform unimodal measures for explaining variance in resistance outcomes.

At present, evidence is insufficient for individual-level prediction or commercial deployment; the appropriate scientific role is mechanism testing under rigorous governance.

5.4. Dual-use and ethical stakes

Research on vulnerability to persuasion is inherently dual-use: the same mechanistic knowledge that could support consumer protection, education, and policy could also sharpen manipulative strategies. Ethical safeguards cannot be limited to informed consent alone, especially in platform-scale or high-stakes settings. They must include transparency about aims and limitations, independent oversight, and

restrictions on exploitative applications—particularly for vulnerable populations (e.g., minors or individuals under severe financial/psychological stress).

6. A Pragmatic Hybrid EEG–fMRI Research Agenda

6.1. Why hybrid EEG–fMRI, and why staged?

Simultaneous EEG–fMRI combines EEG’s temporal sensitivity with fMRI’s spatial specificity and can, in principle, capture rapid markers of regulatory engagement alongside PFC–limbic circuit localization (Mulert & Lemieux, 2010; Warbrick, 2022). However, simultaneous acquisition is technically demanding and vulnerable to artifacts (gradient, ballistocardiogram, motion) and analytic flexibility. Therefore, we recommend a staged agenda: establish robust effects in EEG-only and fMRI-only studies first, then proceed to simultaneous EEG–fMRI for theory-driven questions where the incremental value is justified.

6.2. Stepwise agenda overview

Stage 1 (Unimodal validation):

- EEG-only: identify reliable electrophysiological indices of regulatory engagement during persuasion and resistance.
- fMRI-only: identify robust activation and connectivity contrasts within predefined PFC/ACC–limbic ROIs.

Stage 2 (Simultaneous EEG–fMRI):

- Test EEG–BOLD correspondence and joint predictors of persuasion/resistance.
- Reduce degrees of freedom via pre-registration and constrained ROIs.

Stage 3 (Ecological and interpersonal extensions):

- Incorporate longitudinal stress/digital exposure measures; optionally extend to hyperscanning or group paradigms with careful causal inference.

6.3. Core experimental paradigm for persuasion and resistance

We propose a within-subject paradigm that manipulates message type and regulatory stance:

Message type: emotionally framed persuasive messages vs matched informational controls; short-term vs long-term framing.

Regulatory stance: passive viewing; reappraisal-like regulation (focus on long-term consequences/context); resistance via counter-arguing.

Behavioral outcomes should include trial-level persuasiveness ratings, affect ratings, memory for claims/counterarguments, and incentive-compatible choices when feasible. Sample sizes must be justified with a priori power analyses; underpowered designs are unlikely to yield stable estimates for connectivity or fusion metrics (Button et al., 2013).

6.4. Hypotheses and operational tests

H1 (Connectivity modulation and resistance): Successful resistance to high-arousal messages will be associated with larger condition-dependent modulation of vIPFC/vmPFC–amygdala coupling relative to passive viewing.

H2 (EEG markers of control and PFC/ACC engagement): Increases in frontal midline theta and/or reduced LPP during regulation/resistance will covary with BOLD engagement in ACC and dlPFC.

H3 (Stress and reduced flexibility): Higher chronic stress or allostatic load will be associated with reduced context sensitivity and reduced neural–behavioral coupling, consistent with stress-related PFC/hippocampal vulnerability (McEwen & Gianaros, 2011; McEwen, 2013) and stress-related shifts in control (Schwabe & Wolf, 2011).

6.5. Analysis plan and reproducibility safeguards

fMRI: standard preprocessing with explicit motion modeling; ROI-first inference (PFC subregions, ACC, amygdala, hippocampus, ventral striatum); condition-wise GLMs; gPPI for condition-dependent connectivity; optional dFC within the predefined subnetwork with pre-registered windows/states.

EEG: artifact correction appropriate for simultaneous acquisition; standardized pipelines for gradient and ballistocardiogram correction; ERP (e.g., LPP) and time–frequency analyses (e.g., theta) aligned to message events.

Fusion: EEG-informed fMRI regressors and/or joint decomposition methods (e.g., joint ICA) with out-of-sample validation where possible. Multiple-comparison correction and reporting of effect-size uncertainty are mandatory, given the high degrees of freedom in multimodal analysis (Poldrack et al., 2017).

6.6. Feasibility, quality control, and common failure points

Simultaneous EEG–fMRI is particularly sensitive to artifact and workflow decisions. We recommend using contemporary guidelines from systematic reviews of artifact reduction and technique maturation to inform protocols (Bullock et al., 2021; Warbrick, 2022). Table 2 summarizes common failure points and mitigations.

Table 2. Common failure points in simultaneous EEG–fMRI and minimum mitigations (recommended).

Risk / failure point	Why it matters	Mitigation (minimum)	Reporting requirement
Gradient and ballistocardiogram artifacts (EEG)	Can dominate EEG spectrum/ERPs; especially problematic for frontal channels	Use established GA/BCG correction; record ECG; verify residual artifact spectra; exclude contaminated epochs	Describe correction steps; provide QA plots and residual artifact metrics
Head motion and	Inflates connectivity	Strict motion	Report motion

physiological noise (fMRI)	estimates; biases dFC and PPI results	thresholds; include motion regressors/scrubbing; physiological regressors if available	distributions; sensitivity analyses with/without high-motion subjects
Analytic flexibility in multimodal fusion	Multiple pipelines can yield diverging 'significant' results	Pre-register primary outcomes; restrict ROIs; nested cross-validation for predictive models	Publish code; enumerate all tested models; correct for multiple testing
Small samples / low trial counts	Unstable effect estimates; poor reproducibility	Power analysis; increase N and trials; focus on effect sizes and uncertainty	Justify sample size; report precision and robustness checks

Box 1. Dual-use governance and risk-mitigation checklist (recommended minimum)

Purpose limitation: State explicitly whether the work is for understanding mechanisms, prevention, or policy; avoid optimization-for-influence framing.

Consent and transparency: Use clear participant-facing descriptions; disclose any persuasive content and debrief.

Vulnerability protections: Exclude or add safeguards for minors and high-vulnerability groups; avoid targeting based on stress/mental health indicators.

Data governance: Limit data access; document retention and sharing; prevent downstream commercial transfer of prediction models.

Independent oversight: Involve IRB/ethics boards and, where relevant, external advisory panels for dual-use review.

Reproducibility: Pre-register hypotheses/analysis; share code; report all outcomes; prioritize effect sizes and uncertainty.

Platform-scale research caution: If using large-scale digital experiments, document opt-out/consent and ethical review; explicitly discuss prior controversies (e.g., Verma, 2014).

7. Limitations

First, this is a narrative review; it does not exhaustively enumerate all relevant studies. Second, methodological heterogeneity across tasks, samples, and analytic pipelines limits strong inference. Third, PFC–limbic flexibility is conceptual and requires empirical validation of reliability, discriminant validity, and predictive value. Fourth, simultaneous EEG–fMRI is not universally feasible and increases analytic degrees of freedom; staged development is therefore recommended. Fifth, persuasion-related neuroscience is ethically sensitive and dual-use; governance must be treated as integral to study design rather than an afterthought.

8. Future Directions

Key priorities include: (1) validating alternative operationalizations of PFC–limbic flexibility with test–retest reliability and cross-task generalization; (2) building well-powered unimodal baselines prior to simultaneous EEG–fMRI; (3) developing longitudinal designs integrating stress exposure and digital media use; (4) extending to interpersonal or group paradigms (e.g., hyperscanning) with careful causal inference; and (5) integrating ethics, policy, and governance frameworks that distinguish protective applications from exploitative uses.

9. Conclusion

Cognitive–emotional integration is a central property of how brains regulate behavior in complex environments. Evidence from emotion regulation, stress/allostasis, interpersonal synchrony, and consumer decision-making converges on the importance of flexible, context-sensitive coordination among PFC–limbic circuits. We proposed PFC–limbic flexibility as a falsifiable bridging construct and outlined a pragmatic staged EEG–fMRI agenda for testing hypotheses about persuasion and resistance under strong reproducibility and governance constraints. Progress in understanding influence-related neurobiology should be matched by transparency and ethical safeguards to ensure that scientific advances protect, rather than exploit, human vulnerability.

Supplementary Materials

Dashboard (live demo; exploratory visualization):
<https://amirlotfifar1994-dot.github.io/cv/pfc/index.html>

OSF project file (versioned pointers + offline-view instructions): pfc_dashboard_links.zip (README + SOURCE_URLS.txt; no participant data).

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Competing Interests

The authors declare no competing interests.

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Data and Code Availability

No new empirical datasets or analysis code were generated for this conceptual review.

Dashboard access information is included in the OSF project files associated with this preprint (file: pfc_dashboard_links.zip), and a live demo is available at: <https://amirlotfifar1994-dot.github.io/cv/pfc/index.html>. The dashboard contains no participant data and is provided for exploratory, non-clinical use.

For long-term access, OSF project files should be treated as the version of record for supplementary materials. If an offline dashboard snapshot is later uploaded, it should be archived there as well.

Ethics Statement

Not applicable. This manuscript is a conceptual review and did not involve the collection of new data from human or animal participants. The optional dashboard contains no participant data.

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